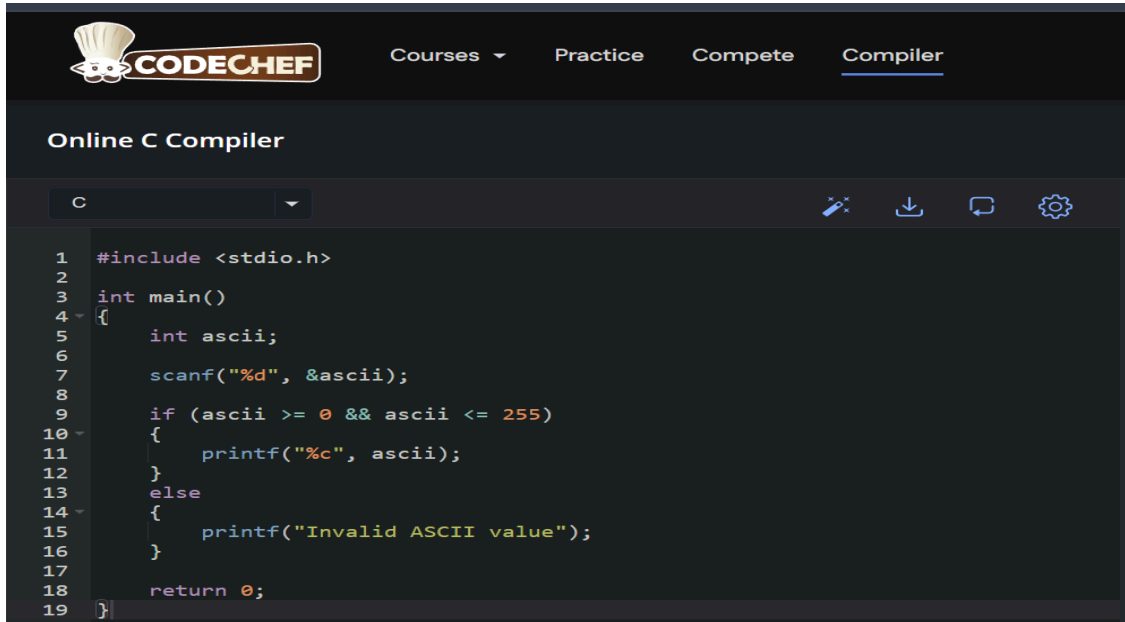


Q.1 Write a C program to accept an integer value (ASCII) from the user and print it's equivalent character. The ASCII values vary from 0 to 255.

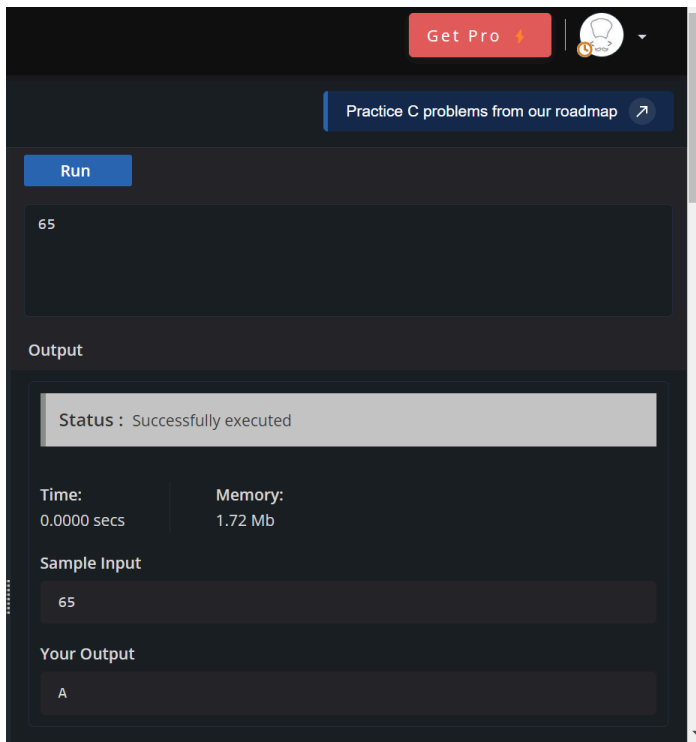
CODE:



The screenshot shows the CodeChef Online C Compiler interface. The language is set to C. The code in the editor is as follows:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int ascii;
6
7      scanf("%d", &ascii);
8
9      if (ascii >= 0 && ascii <= 255)
10     {
11         printf("%c", ascii);
12     }
13     else
14     {
15         printf("Invalid ASCII value");
16     }
17
18     return 0;
19 }
```

OUTPUT:



The screenshot shows the output of the program. The input value is 65, and the output is the character 'A'. The status is "Successfully executed".

Run

65

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.72 Mb

Sample Input

65

Your Output

A

Q.2 Write a program to swap the value of two variable for example: -

value before swapping

a=10

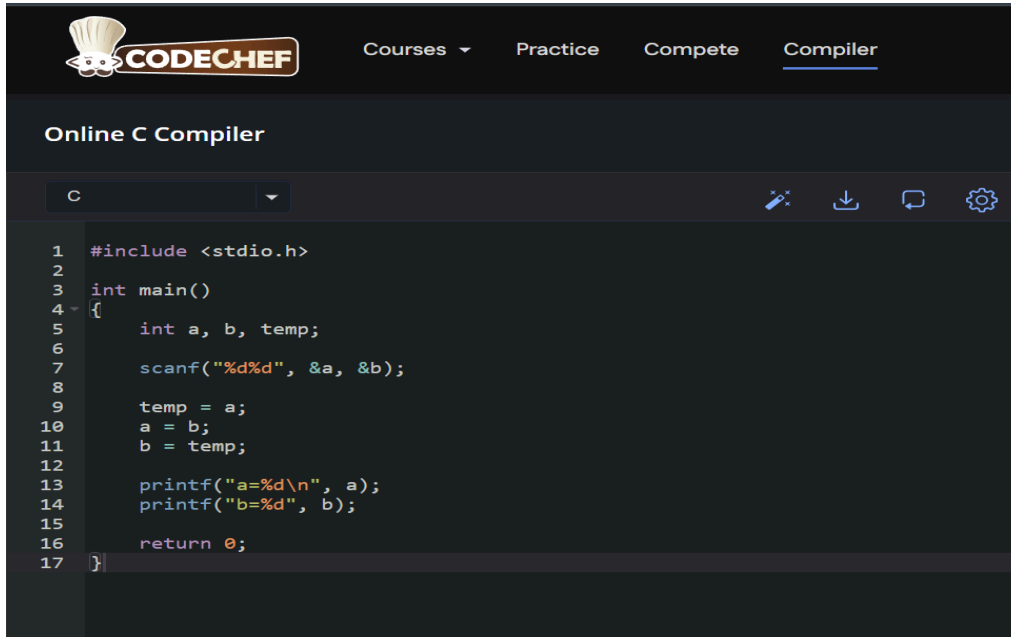
b=20

value after swapping

a=20

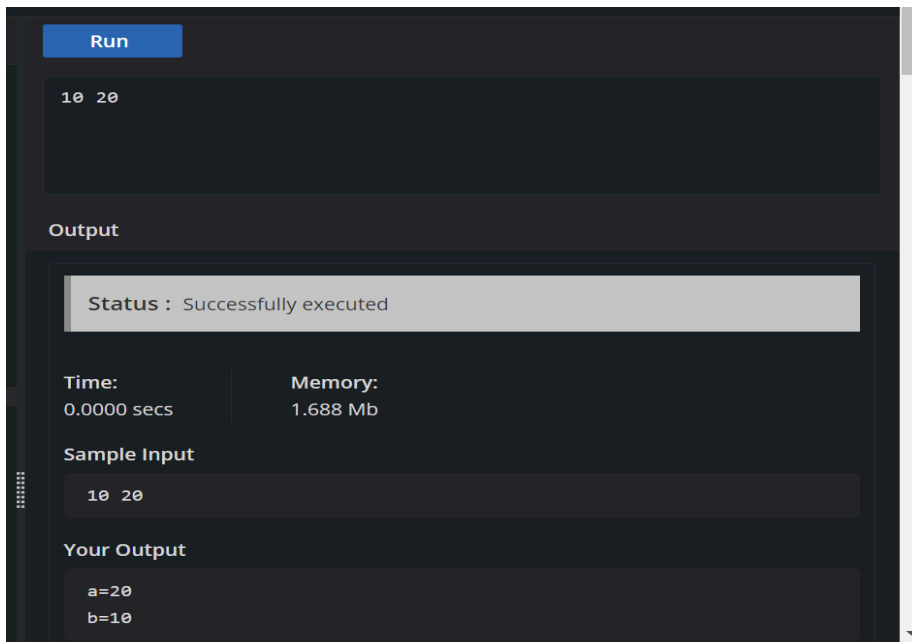
b=10

CODE:



```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a, b, temp;
6
7     scanf("%d%d", &a, &b);
8
9     temp = a;
10    a = b;
11    b = temp;
12
13    printf("a=%d\n", a);
14    printf("b=%d", b);
15
16    return 0;
17 }
```

OUTPUT:



Run

10 20

Output

Status : Successfully executed

Time:	Memory:
0.0000 secs	1.688 Mb

Sample Input

10 20

Your Output

a=20
b=10

Q.3 Write a C program that takes a point (x, y) and displays the quadrant in which it falls using if-else.

CODE:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int x, y;
6
7      scanf("%d%d", &x, &y);
8
9      if (x > 0 && y > 0)
10     {
11         printf("First Quadrant");
12     }
13     else if (x < 0 && y > 0)
14     {
15         printf("Second Quadrant");
16     }
17     else if (x < 0 && y < 0)
18     {
19         printf("Third Quadrant");
20     }
21     else if (x > 0 && y < 0)
22     {
23         printf("Fourth Quadrant");
24     }
25     else if (x == 0 && y == 0)
26     {
27         printf("Origin");
28     }
29     else if (x == 0)
30     {
31         printf("Point lies on Y-axis");
32     }
33     else if (y == 0)
34     {
35         printf("Point lies on X-axis");
36     }
37
38     return 0;
39 }
```

OUTPUT:

The screenshot shows a code execution environment. At the top, the input '4 5' is entered. Below it, the 'Output' section displays 'First Quadrant'. Above the output, a status bar indicates 'Status : Successfully executed'. Below the status bar, performance metrics are shown: 'Time: 0.0000 secs' and 'Memory: 1.688 Mb'. At the bottom, the 'Sample Input' section shows '4 5' and the 'Your Output' section shows 'First Quadrant'.

4 5

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.688 Mb

Sample Input

4 5

Your Output

First Quadrant

Q.4 Write a C program to calculate the division of a student in 5 subjects. Consider the maximum marks in a subject is 100.

The division will be decided as below

Percentage	Division
>80	Distinction
>=60	First
>=45	Second
<45	Fail

CODE:

```
Online C Compiler
C
1 #include <stdio.h>
2
3 int main()
4 {
5     int m1, m2, m3, m4, m5;
6     int total;
7     float percentage;
8
9     scanf("%d%d%d%d%d", &m1, &m2, &m3, &m4, &m5);
10
11     total = m1 + m2 + m3 + m4 + m5;
12     percentage = total / 5.0;
13
14     printf("Total = %d\n", total);
15     printf("Percentage = %.2f\n", percentage);
16
17     if (percentage > 80)
18     {
19         printf("Division = Distinction");
20     }
21     else if (percentage >= 60)
22     {
23         printf("Division = First");
24     }
25     else if (percentage >= 45)
26     {
27         printf("Division = Second");
28     }
29     else
30     {
31         printf("Division = Fail");
32     }
33
34     return 0;
35 }
```

OUTPUT:

Run

85 78 90 88 80

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.78 Mb

Sample Input

85 78 90 88 80

Your Output

Total = 421
Percentage = 84.20
Division = Distinction

Q.5 Write a program to check if a character is a consonant or a vowel.

CODE:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      char ch;
6
7      scanf("%c", &ch);
8
9      if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
10     {
11         if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
12             ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U')
13         {
14             printf("Vowel");
15         }
16         else
17         {
18             printf("Consonant");
19         }
20     }
21     else
22     {
23         printf("Invalid character");
24     }
25
26     return 0;
27 }
```

OUTPUT:

Run

e

Output

Status : Successfully executed

Time:
0.0000 secs

Memory:
1.568 Mb

Sample Input

e

Your Output

Vowel